

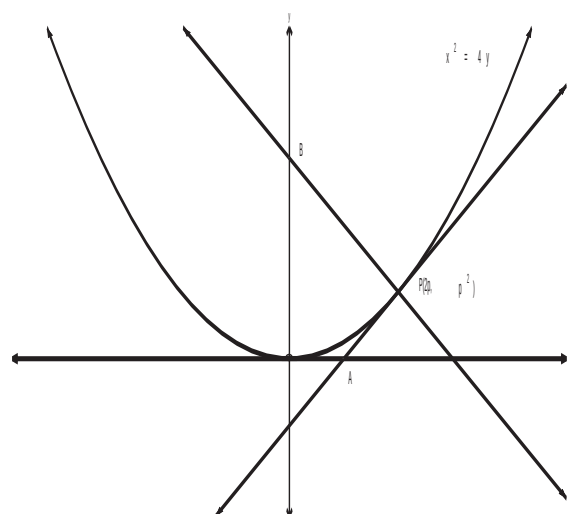
Total marks – 75**Attempt Questions 1-4**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

QUESTION 1	(22 marks)	START A NEW BOOKLET	Marks
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- | | | |
|------|--|---|
| (a) | Evaluate $\int \frac{\cos x}{\sin x} dx$. | 2 |
| (b) | Consider the curves $y = e^x$ and $y = x^3 - 3x + 1$. | |
| (i) | Show that they have a common point at (0,1). | 1 |
| (ii) | Find the acute angle (to the nearest minute) between the tangents to the curves at this point. | 3 |
| (c) | Use $2\cos^2 \theta = 1 + \cos 2\theta$ to prove that $\cos \frac{\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}$. | 3 |
| (d) | Let $f(x) = \ln(\tan x)$, $0 < x < \frac{\pi}{2}$. Show that $f'(x) = 2\operatorname{cosec} 2x$. | 2 |
| (e) | Use mathematical induction to prove that $3^{2n-1} + 5$ is divisible by 8, for all integers $n \geq 1$. | 5 |
| (f) | The diagram shows a point $P(2p, p^2)$ on the parabola $x^2 = 4y$. The tangent to the parabola at P cuts the x-axis at A. The normal to the parabola at point P cuts the y-axis at B. | |

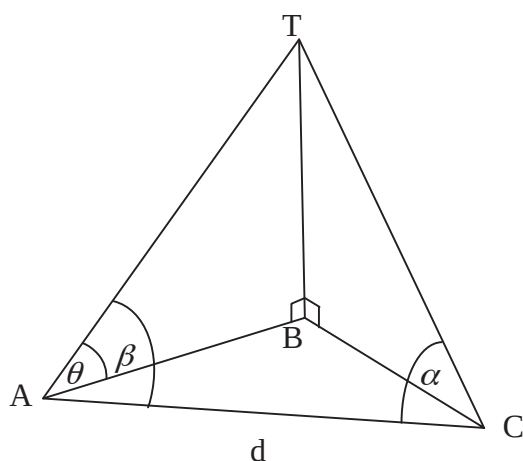
- | | | |
|-------|---|---|
| (i) | Find the equation of the tangent AP. | 1 |
| (ii) | Show that B has coordinate $(0, p^2 + 2)$ | 2 |
| (iii) | Let M be the midpoint of AB. Find the Cartesian equation of the locus of M. | 3 |



QUESTION 2	(18 marks)	START A NEW BOOKLET	Marks
(a)	Evaluate $\int_0^1 \frac{1}{\sqrt{4-x^2}} dx$		2
(b)	(i) Differentiate $\sin^{-1} x^3$ with respect to x .		2
	(ii) Hence or otherwise find $\int \frac{2x^2}{\sqrt{1-x^6}} dx$		1
(c)	Consider the function $f(x) = 3\sin^{-1}(2x-1)$		
	(i) Find the domain of $f(x)$		2
	(ii) Find the range of $f(x)$		2
	(iii) Sketch the graph of $f(x)$		2
(d)	Consider the function $f(x) = e^{-x} - e^x$		
	(i) Show that $f(x)$ is decreasing for all values of x .		2
	(ii) Show that the inverse function is given by:		
	$f^{-1}(x) = \log_e \left(\frac{-x + \sqrt{x^2 + 4}}{2} \right)$		3
	(iii) Hence solve $e^{-x} - e^x = 6$, giving your answer in simplest surd form.		2

QUESTION 3 (18 marks) **START A NEW BOOKLET** **Marks**

- (a) Solve $\sin^2 \theta = \frac{1}{2} \sin 2\theta$, $0 \leq \theta \leq 2\pi$ **3**
- (b) (i) Express $y = \sqrt{3} \sin x + \cos x$ in the form $R \sin(x + \alpha)$, $R > 0$ and α is acute. **3**
- (ii) Find the maximum and minimum values of $y = \sqrt{3} \sin x + \cos x$ **1**
- (iii) Solve $\sqrt{3} \sin x + \cos x = 1$, $0 \leq x \leq 2\pi$ **2**
- (c) Evaluate $\int_0^{\frac{\pi}{4}} \sin^2\left(\frac{1}{2}x\right) dx$ **3**
- (d) Using the substitution $t = \tan \frac{\theta}{2}$ prove that $\frac{1 - \cos \theta}{1 + \cos \theta} = \tan^2 \frac{\theta}{2}$ **3**
- (e) TB is a tower such that the angle of elevation of T, the top of the tower, from point A is θ . A, B and C are on the same level and $\angle CAT = \beta$ and $\angle ACT = \alpha$. If $AC = d$ metres then show that the height of the tower is given by: $TB = \frac{d \sin \theta \sin \alpha}{\sin(\alpha + \beta)}$. **3**



QUESTION 4 (17 marks) **START A NEW BOOKLET** **Marks**

- (a) The velocity v , in metres per second, of a particle travelling in a straight line is given by $v = 2t - 4$. The initial displacement of the particle is given by $x = 3$.
- (i) Find the acceleration of the particle. **1**
 - (ii) Find the displacement of the particle when it is stationary. **3**
 - (iii) How far does the particle travel in the first 3 seconds. **1**
- (b) A bug is oscillating in simple harmonic motion such that its displacement x metres from a fixed point O at time t seconds is given by the equation ~~$x = -4x$~~ $x = -4x$. When $t = 0, v = 2\text{ m/s}$ and $x = 5$.
- (i) Show that $x = a \cos(2t + \alpha)$ is a solution for this equation, where a and α are constants. **1**
 - (ii) Find the period of the motion. **1**
 - (iii) Show that the amplitude of the oscillation is $\sqrt{26}$. **3**
 - (iv) What is the maximum speed of the bug? **2**
- (c) The acceleration of a particle P is given by the equation
- $$\frac{d^2x}{dt^2} = 8x(x^2 + 4)$$
- where x metres is the displacement of P from a fixed point O after t seconds.
- Initially the particle is at O and has velocity 8 ms^{-1} in the positive direction.
- (i) Show that the speed at any position x is given by $2(x^2 + 4)$ **3**
 - (ii) Hence find the time taken for the particle to travel 2 metres from O. **2**

END OF THE PAPER

(a) $\int \frac{\cos x}{\sin x} dx = \ln \sin x + C$ ✓✓

(b) (i) $y = e^x : x=0, y=1$
 $y = x^3 - 3x + 1 : x=0, y=1$ ✓

(ii) $m_1 = e^0 = 1$
 $m_2 = 3x^2 - 3 = -3$ ✓

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{1 + 3}{1 - 3} \right|$$

$$= \left| \frac{4}{-2} \right|$$

$$= 2$$

$\therefore \theta = \tan^{-1} 2$
 $= 63^\circ 26'$ (nearest minute) ✓

(c) $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$

$$\cos \theta = \pm \sqrt{\frac{1 + \cos 2\theta}{2}}$$
 ✓

$\therefore \cos \frac{\pi}{8} = \sqrt{\frac{1 + \cos \frac{\pi}{4}}{2}}$ ($\cos \frac{\pi}{8} > 0$) ✓

$$= \sqrt{\frac{1 + \sqrt{2}/2}{2}}$$

$$= \sqrt{\frac{2 + \sqrt{2}}{4}}$$

$$= \frac{\sqrt{2 + \sqrt{2}}}{2}$$
 ✓

Must discuss
+ and -

abd	8
c	3
ef	11

(d) $f(x) = \ln \tan x$ $0 < x < \frac{\pi}{2}$

$$f'(x) = \frac{\sec^2 x}{\tan x}$$

$$= \frac{1}{\sin x \cos x}$$

$$= \frac{2}{\sin 2x}$$

$$= 2 \operatorname{cosec} 2x$$

(e) To prove: $3^{2n-1} + 5$ is divis. by 8 for all integers $n \geq 1$

$n=1$: $3 + 5 = 8$

$\therefore$ true for $n=1$

$n=k$: assume that it is true for $n=k$

$$3^{2k-1} + 5 = 8M \quad (M \in \mathbb{Z})$$

$$\therefore 3^{2k-1} = 8M - 5$$

$n=k+1$: prove true for $n=k+1$

$$\text{i.e. } 3^{2(k+1)-1} + 5 = 8N \quad (N \in \mathbb{Z})$$

$$\text{LHS} = 3^{2k+1} + 5$$

$$= 3^2 \cdot 3^{2k-1} + 5$$

$$= 9(8M - 5) + 5$$

$$= 72M - 40$$

$$= 8(9M - 5)$$

$$= 8N$$

(since $M \in \mathbb{Z}$ then $9M - 5 \in \mathbb{Z}$)

$\therefore$ Since it is true for $n=1$, and if we assume that it is true for $n=k$ then it is also true for $n=k+1$, then by the principle of math. induction it is true for $n=1, 2, 3, \dots$

$$(f)(i) \quad y - y_1 = m(x - x_1)$$

$$y - p^2 = p(x - 2p)$$

$$y - p^2 = px - 2p^2$$

$$px - y - p^2 = 0$$

$$(ii) \quad \text{gradient of normal is } -\frac{1}{p}$$

$$y - p^2 = -\frac{1}{p}(x - 2p)$$

$$y - p^2 = -\frac{1}{p}x + 2$$

$$y = -\frac{1}{p}x + 2 + p^2$$

$$\text{when } x=0 \quad y = p^2 + 2$$

$$\therefore B \text{ is } (0, p^2 + 2)$$

$$(iii) \quad \text{in } px - y - p^2 = 0 \quad \text{let } y = 0$$

$$px - p^2 = 0$$

$$px = p^2$$

$$x = p$$

$$\therefore A \text{ is } (p, 0)$$

$$\therefore M = \left(\frac{0+p}{2}, \frac{p^2+2+0}{2} \right)$$

$$= \left(\frac{p}{2}, \frac{p^2+2}{2} \right)$$

$$\text{let } x = \frac{p}{2} \quad \therefore p = 2x$$

$$y = \frac{p^2+2}{2}$$

$$= \frac{(2x)^2+2}{2}$$

$$= \frac{4x^2+2}{2}$$

$$\therefore y = 2x^2 + 1$$

$$2(a) \int_0^1 \frac{1}{\sqrt{1-x^2}} dx = \left[\sin^{-1} x \right]_0^1 \checkmark$$

$$= \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{6} \checkmark$$

$$(b) (i) \frac{d[\sin^{-1} x^3]}{dx} = \frac{3x^2}{\sqrt{1-x^6}} \checkmark$$

$$(ii) \int \frac{2x^2}{\sqrt{1-x^6}} dx = \frac{2}{3} \sin^{-1} x^3 + c \checkmark$$

$$(c) f(x) = 3 \sin^{-1}(2x-1)$$

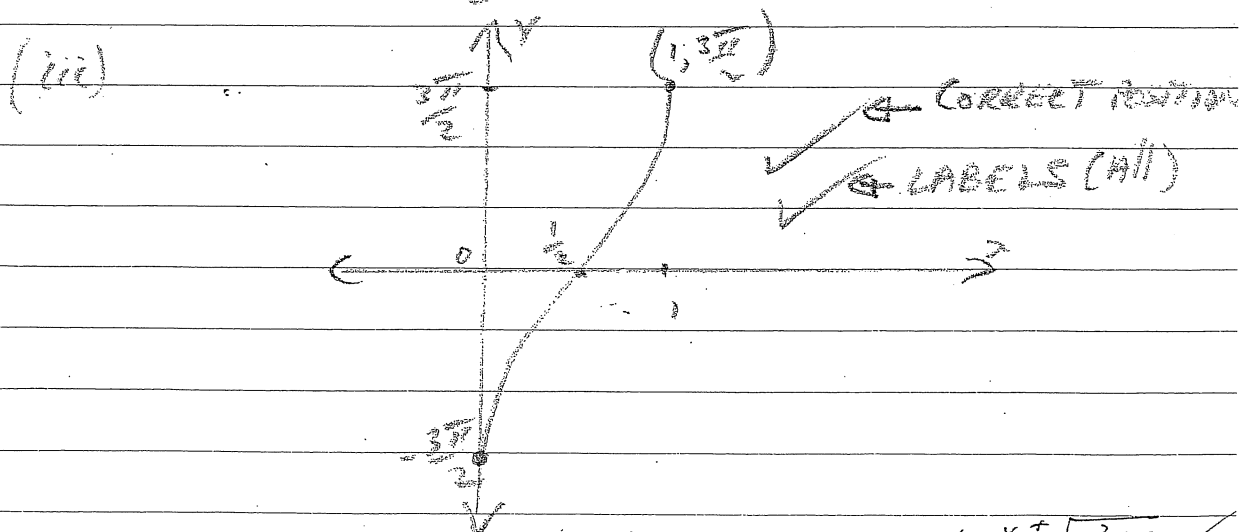
$$(i) \text{ Domain } -1 \leq 2x-1 \leq 1 \checkmark$$

$$0 \leq 2x \leq 2$$

$$0 \leq x \leq 1 \checkmark$$

$$(ii) \text{ Range } -\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2} \checkmark$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2} \checkmark$$



$$(d)(i) f(x) = e^{-x} - e^x$$

$$f'(x) = -e^{-x} - e^x$$

$$= -(e^{-x} + e^x) \checkmark$$

$\neq 0$ as $e^{-x} > 0$ and $e^x > 0$ for all x ✓

$$(ii) \text{ let } x = e^{-y} - e^y$$

$$\therefore e^y x = 1 - e^{2y}$$

$$\therefore (e^y)^2 + x(e^y) - 1 = 0$$

$$\therefore e^y = \frac{-x \pm \sqrt{x^2 + 4}}{2} \checkmark$$

[TAK +]

$$y = \ln \left[\frac{-x + \sqrt{x^2 + 4}}{2} \right] \checkmark$$

$$(iii) \text{ let } f[f(x)] = x$$

$$\therefore x = \ln \left[\frac{-6 + \sqrt{40}}{2} \right] \checkmark$$

$$= \ln[-3 + \sqrt{10}] \checkmark$$

Solutions

Q3

$$(a) \quad \sin^2 \theta = \frac{1}{2} \sin 2\theta, \quad 0 \leq \theta \leq 2\pi.$$

$$\sin^2 \theta = \frac{1}{2} (2 \sin \theta \cos \theta) \quad |$$

$$\therefore \sin^2 \theta - \sin \theta \cos \theta = 0.$$

$$\therefore \sin \theta (\sin \theta - \cos \theta) = 0$$

$$\sin \theta = 0 \quad \text{or} \quad \sin \theta - \cos \theta = 0$$

$$\sin \theta = \cos \theta; \quad \cos \theta \neq 0.$$

$$\therefore \theta = 0, \pi, 2\pi \quad \therefore \tan \theta = 1$$

$$| \quad \text{or} \quad \theta = \pi/4, 5\pi/4. \quad |$$

$$\therefore \theta = 0, \pi/4, 5\pi/4, \pi, 2\pi. \quad (3)$$

$$(b) (i) \quad y = \sqrt{3} \sin x + \cos x$$

$$\text{let } \sqrt{3} \sin x + \cos x = R \sin(x + \alpha)$$

$$\therefore \sqrt{3} \sin x + \cos x = R \sin x \cos \alpha + R \cos x \sin \alpha.$$

$$\therefore \left. \begin{aligned} R \cos \alpha &= \sqrt{3} \\ R \sin \alpha &= 1 \end{aligned} \right\} |$$

$$R^2 = 4 \quad \therefore R = 2$$

$$\text{and } \tan \alpha = \frac{1}{\sqrt{3}} \quad \therefore \alpha = \pi/6. \quad (3)$$

$$\therefore \sqrt{3} \sin x + \cos x = 2 \sin(x + \pi/6).$$

$$(ii) \quad \text{Maximum + Minimum} : \quad -2 \leq 2 \sin(x + \pi/6) \leq 2.$$

$$\therefore \text{Maximum} = 2 \quad \text{and} \quad \text{Minimum} = -2. \quad (1)$$

$$(iii) \quad \text{If } \sqrt{3} \sin x + \cos x = 1; \quad 0 \leq x \leq 2\pi.$$

$$\text{then } 2 \sin(x + \pi/6) = 1; \quad \pi/6 \leq x + \pi/6 \leq 13\pi/6$$

$$\therefore \sin(x + \pi/6) = \frac{1}{2}$$

$$\text{If } \sin \alpha = \frac{1}{2}, \quad \alpha = \pi/6.$$

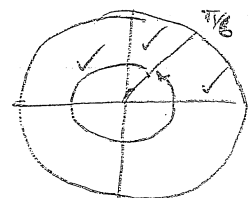
$$\therefore x + \pi/6 = \pi/6 \quad \text{or} \quad x + \pi/6 = 5\pi/6$$

$$\therefore x = -\pi/6 \quad \text{or} \quad x = 2\pi/3$$

invalid

$$\text{or } x = \pi/6, 2\pi$$

$$\therefore x = \pi/6, 2\pi/3, 5\pi/3, 2\pi. \quad (2)$$



$$\text{or } x + \pi/6 = 13\pi/6.$$

Q3 (c) $\int_0^{\pi/4} \sin^2\left(\frac{1}{2}x\right) dx$; let $\sin^2\left(\frac{x}{2}\right) = \frac{1 - \cos x}{2}$

Now, $\sin 2\theta = 1 - 2\sin^2\theta$
 $\therefore 2\sin^2\theta = 1 - \sin 2\theta$
 $\therefore \sin^2\theta = \frac{1}{2}(1 - \sin 2\theta)$
 $\therefore \sin^2\left(\frac{1}{2}x\right) = \frac{1}{2}(1 - \sin x)$

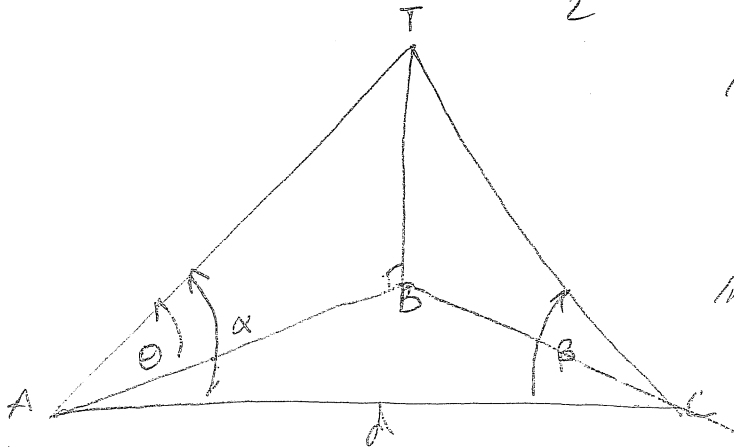
$$\begin{aligned} \therefore \int_0^{\pi/4} \sin^2\left(\frac{1}{2}x\right) dx &= \frac{1}{2} \int_0^{\pi/4} (1 - \sin x) dx \\ &= \frac{1}{2} \left[x + \cos x \right]_0^{\pi/4} \\ &= \frac{1}{2} \left(\frac{\pi}{4} + \frac{\sqrt{2}}{2} \right) - \frac{1}{2} (0) \\ &= \frac{1}{2} \left(\frac{\pi}{4} + \frac{\sqrt{2}}{2} \right) \quad (3) \end{aligned}$$

(d) Let $t = \tan \frac{\theta}{2}$; $\frac{1 - \cos \theta}{1 + \cos \theta} = \tan^2 \frac{\theta}{2}$

Now $\cos \theta = \frac{1 - t^2}{1 + t^2}$

$$\begin{aligned} \therefore \frac{1 - \cos \theta}{1 + \cos \theta} &= \left(\frac{1 - \frac{1 - t^2}{1 + t^2}}{1 + \frac{1 - t^2}{1 + t^2}} \right) = \left(\frac{1 + \frac{1 - t^2}{1 + t^2}}{1 + \frac{1 - t^2}{1 + t^2}} \right) \\ &= \left(\frac{1 + t^2 - 1 + t^2}{1 + t^2} \right) \div \left(\frac{1 + t^2 + 1 - t^2}{1 + t^2} \right) \\ &= \frac{2t^2}{1 + t^2} \times \frac{1 + t^2}{2} \\ &= t^2 \\ &= \tan^2 \frac{\theta}{2} \quad (3) \end{aligned}$$

(e)



In $\triangle ABT$
 $\sin \theta = \frac{TB}{AT}$

$\therefore TB = AT \sin \theta$

In $\triangle CAT$, $\angle ATC = [180 - (\alpha + \beta)]$

$\therefore \frac{AT}{\sin \beta} = \frac{AC}{\sin (180 - (\alpha + \beta))}$

$\therefore AT = \frac{AC \sin \beta}{\sin (\alpha + \beta)}$ (3)

$\therefore AT = \frac{d \sin \beta}{\sin (\alpha + \beta)}$; $TB = \frac{d \sin \beta \sin \theta}{\sin (\alpha + \beta)}$

QUESTION FOUR

a) $v = 2t - 4$

(i) $a = 2 \text{ m/s}^2$

(ii) stationary when $t = 2$

$$x = t^2 - 4t + c$$

when $t = 0$ $x = 3 \therefore c = 3$

$$x = t^2 - 4t + 3$$

when $t = 2$ $x = 4 - 8 + 3$
 $= -1 \text{ m}$

(iii) $\left. \begin{array}{l} \text{when } t = 0 \quad x = 3 \\ \text{when } t = 2 \quad x = -1 \\ \text{when } t = 3 \quad x = 0 \end{array} \right\} \begin{array}{l} 4 \text{ m} \\ 1 \text{ m} \end{array}$

particle travels 5m in the first 3 seconds.

5

b) $\ddot{x} = -4x$, when $t = 0$ $v = 2$ and $x = 5$

(i) $x = a \cos(2t + \alpha)$

$$\dot{x} = -2a \sin(2t + \alpha)$$

$$\ddot{x} = -4a \cos(2t + \alpha)$$

$$= -4 \left[a \cos(2t + \alpha) \right]$$

$$= -4x$$

(ii) $T = \frac{2\pi}{2} = \pi$

(iii) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -4x$

$$\therefore \frac{1}{2} v^2 = -2x^2 + c$$

when $v = 2$, $x = 5$

$$2 = -50 + c$$

$$\therefore c = 52$$

$$\therefore v^2 = -4x^2 + 104$$

$$v^2 = 4(26 - x^2)$$

$$v^2 = 4(a^2 - x^2)$$

$$\therefore a = \sqrt{26}$$

(iv) $\dot{x} = -2a \sin(2t + \alpha)$

$$x = -2\sqrt{26} \sin(2t + \alpha)$$

$$-1 \leq \sin(2t + \alpha) \leq 1$$

$$\therefore \text{max speed is } 2\sqrt{26} \text{ m/s}$$

5

(c) $\ddot{x} = 8x(x^2 + 4) = 8x^3 + 32x$

when $t = 0$, $x = 0$ and $v = 8$

(i) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 8x^3 + 32x$

$$\frac{1}{2} v^2 = 2x^4 + 16x^2 + c$$

when $x = 0$, $v = 8 \therefore c = 32$

$$\therefore \frac{1}{2} v^2 = 2x^4 + 16x^2 + 32$$

$$v^2 = 4x^4 + 32x^2 + 64$$

$$= 4(x^4 + 8x^2 + 16)$$

$$= 4(x^2 + 4)^2$$

$$\therefore \text{speed} = \sqrt{v^2} = 2(x^2 + 4)$$

(ii) $\frac{dx}{dt} = \frac{1}{2(x^2 + 4)}$

Note v is always positive. Particle starts with $v > 0$ and $\ddot{x} > 0 \Rightarrow v \neq 0$

$$\therefore t = \frac{1}{4} \tan^{-1} \frac{x}{2} + c$$

when $t = 0$, $x = 0 \therefore c = 0$

$$\therefore t = \frac{1}{4} \tan^{-1} \frac{x}{2}$$

when $x = 2$, $t = \frac{1}{4} \tan^{-1} 1$
 $= \frac{\pi}{16} \text{ seconds}$

5

17

Alternative solution to (b) (ii)

$$x = a \cos(2t + \alpha)$$

$$\dot{x} = -2a \sin(2t + \alpha)$$

$$\text{when } t=0 \quad x=5 \text{ and } \dot{x}=-2$$

$$5 = a \cos \alpha \text{ and } -2 = -2a \sin \alpha$$

$$\cos \alpha = \frac{5}{a}, \quad \sin \alpha = \frac{-1}{a}$$

$$\therefore \frac{25}{a^2} + \frac{1}{a^2} = 1$$

$$\therefore 26 = a^2$$

$$\therefore a = \sqrt{26}$$